Chapter 10. The Lebesgue Integral

Lebesgue measure, measurable functions, convergence theorems, L1 and L2, Riesz–Fischer

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10.1 Measurable sets and Lebesgue measure

Proposition 10.1 (10.1.1)

Every open $U \subset \mathbb{R}$ is a union of countably many disjoint open intervals, and this representation is unique.

Proof. For $x \in U$ let $I_x = (a_x, b_x)$ with $a_x = \inf\{a : (a, x] \subset U\}$, $b_x = \sup\{b : [x, b) \subset U\}$; $I_x \subset U$ is the largest open interval in U containing x. If $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval in U, so $I_x = I_y$. Each I_x contains a rational, and distinct ones are disjoint, so there are countably many. The intervals are the components of U, which gives uniqueness. ■

For $U = \bigsqcup(a_n, b_n)$ put $\lambda(U) = \sum(b_n - a_n)$ (independent of the order, Proposition 2.16 of Volume I), and for any $S \subset \mathbb{R}$ define the **outer measure**

$$\mu(S) = \inf\{\lambda(U) : U \supset S \text{ open}\} \in [0, \infty].$$

Proposition 10.2 (10.1.2)

(a) $S \subset T \Rightarrow \mu(S) \leq \mu(T)$.
(b) If I is an interval with end points $a \leq b$, then $\mu(I) = b - a$.
(c) $\mu(x + S) = \mu(S)$.
(d) $\mu(\bigcup S_n) \leq \sum \mu(S_n)$ for countably many S_n.

Proof. First, if an open interval (a, b) is covered by open intervals J_k, then $b - a \leq \sum |J_k|$: for $a < c < d < b$ finitely many J_k cover the compact $[c, d]$, and by choosing a chain of overlapping intervals from c to d, $d - c \leq \sum |J_k|$; let $c \to a$, $d \to b$. Summing over components (Proposition 6.9), $\lambda(U) \leq \sum \lambda(U_n)$ whenever $U \subset \bigcup U_n$ are open.

(a) is immediate.

(b) For $I = [a, b]$, $\mu(I) \leq \lambda((a - \epsilon, b + \epsilon)) = b - a + 2\epsilon$; and an open $U \supset I$ has a component containing $[a, b]$, so $\lambda(U) \geq b - a$. Other intervals are similar.

(c) Translation preserves λ of open sets.

(d) We may assume all $\mu(S_n) < \infty$. Choose open $U_n \supset S_n$ with $\lambda(U_n) < \mu(S_n) + \epsilon 2^{-n}$; then $\mu(\bigcup S_n) \leq \lambda(\bigcup U_n) \leq \sum \lambda(U_n) < \sum \mu(S_n) + \epsilon$. ■

In particular countable sets have outer measure 0, so $[0, 1]$ is uncountable. A set $E \subset \mathbb{R}$ is **measurable** if

$$\mu(X) = \mu(X \cap E) + \mu(X \cap E^c) \qquad \text{for every } X \subset \mathbb{R};$$

let $\mathfrak{M}$ be the family of measurable sets. By (d), only $\geq$ needs checking, and only for $\mu(X) < \infty$.

Theorem 10.3 (10.1.3)

(a) $\emptyset \in \mathfrak{M}$.
(b) $E \in \mathfrak{M} \Rightarrow E^c \in \mathfrak{M}$.
(c) $E, F \in \mathfrak{M} \Rightarrow E \cup F \in \mathfrak{M}$.
(d) $E_n \in \mathfrak{M} \Rightarrow \bigcup E_n \in \mathfrak{M}$.
(e) If $\mu(E) = 0$, then $E \in \mathfrak{M}$.
(f) Every interval is measurable.
(g) Every open set is measurable.

Proof. (a), (b) are clear from the symmetric definition.

(c) Since $X \cap (E \cup F) = (X \cap E) \cup (X \cap F \cap E^c)$,

$$\mu(X\cap(E\cup F))+\mu(X\cap(E\cup F)^c) \le \mu(X\cap E)+\mu(X\cap E^c\cap F)+\mu(X\cap E^c\cap F^c) = \mu(X\cap E)+\mu(X\cap E^c) = \mu(X),$$

using $F \in \mathfrak{M}$ (for the set $X \cap E^c$) and then $E \in \mathfrak{M}$.

(d) Let $F_n = E_n - (E_1 \cup \cdots \cup E_{n-1})$ (measurable by (b), (c), as $A \cap B = (A^c \cup B^c)^c$), disjoint, and $G_n = F_1 \cup \cdots \cup F_n$, $E = \bigcup E_n = \bigsqcup F_n$. Using $F_n \in \mathfrak{M}$ for the set $X \cap G_n$, $\mu(X \cap G_n) = \mu(X \cap G_{n-1}) + \mu(X \cap F_n)$, so $\mu(X \cap G_n) = \sum_{i\le n} \mu(X \cap F_i)$. As $G_n \in \mathfrak{M}$ and $G_n^c \supset E^c$,

$$\mu(X) = \mu(X \cap G_n) + \mu(X \cap G_n^c) \ge \sum_{i\le n} \mu(X \cap F_i) + \mu(X \cap E^c).$$

Let $n \to \infty$ and use (d) of Proposition 10.2: $\mu(X) \ge \mu(X \cap E) + \mu(X \cap E^c)$.

(e) $\mu(X \cap E) = 0$ and $\mu(X \cap E^c) \le \mu(X)$.

(f) By (b), (d) and (e) it suffices to treat $I = (a, \infty)$. Given X and ϵ, take open $U \supset X$ with $\lambda(U) < \mu(X)+\epsilon$. The open sets $U\cap(a,\infty)$, $U\cap(-\infty,a)$ satisfy $\lambda(U\cap(a,\infty))+\lambda(U\cap(-\infty,a)) \le \lambda(U)$ (each component of U is split into two intervals), so $\mu(X \cap I) + \mu(X \cap I^c) \le \lambda(U \cap (a,\infty)) + \lambda(U \cap (-\infty, a)) + \mu(\{a\}) < \mu(X) + \epsilon$.

(g) Proposition 10.1, (f) and (d). ■

So $\mathfrak{M}$ is a σ-algebra containing all Borel sets. The **Lebesgue measure** is $m = \mu|_{\mathfrak{M}}$.

Theorem 10.4 (10.1.4, countable additivity)

If $E_n \in \mathfrak{M}$ are disjoint, then $m(\bigsqcup E_n) = \sum m(E_n)$.

Proof. In the proof of Theorem 10.3(d) take $X = E = \bigsqcup E_n$ (so $F_n = E_n$): $m(E) \ge \sum m(E_n)$; the reverse is Proposition 10.2(d). ■

Consequences: if $E_1 \subset E_2 \subset \cdots$, then $m(\bigcup E_n) = \lim m(E_n)$ (write the union as $\bigsqcup(E_n - E_{n-1})$); if $E_1 \supset E_2 \supset \cdots$ and $m(E_1) < \infty$, then $m(\bigcap E_n) = \lim m(E_n)$. By definition of μ, every measurable E satisfies $m(E) = \inf\{m(U) : U \supset E \text{ open}\}$, and for bounded E, $m(E) = \sup\{m(K) : K \subset E \text{ compact}\}$ (apply the outer approximation to $[a,b] - E$). Not every set is measurable (Vitali): choosing one point in $[0,1)$ from each coset of $\mathbb{Q}$ gives a set V whose rational translates mod 1 are disjoint and cover $[0,1)$, so $m(V)$ could be neither 0 nor positive.

10.2 Measurable functions and the integral

Let $E \in \mathfrak{M}$ and $\mathbb{R}^* = [-\infty, \infty]$. $f : E \to \mathbb{R}^*$ is **measurable** if $\{x \in E : f(x) > a\} \in \mathfrak{M}$ for every $a \in \mathbb{R}$; a complex function is measurable if its real and imaginary parts are.

Proposition 10.5 (10.2.1)

For $f : E \to \mathbb{R}^*$ the following are equivalent:
(a) f is measurable;
(b) $\{f \ge a\} \in \mathfrak{M}$ for all a;
(c) $\{f < a\} \in \mathfrak{M}$ for all a;
(d) $\{f \le a\} \in \mathfrak{M}$ for all a.

Proof. $\{f \ge a\} = \bigcap_n \{f > a - 1/n\}$, $\{f > a\} = \bigcup_n \{f \ge a + 1/n\}$, and (c), (d) are the complements (relative to E) of (b), (a). ■

Corollary 10.6 (10.2.2)

If f is measurable, then $\{f = a\}$ is measurable for every $a \in \mathbb{R}^*$.

Proof. $\{f = a\} = \{f \geq a\} \cap \{f \leq a\}$ for real a, and $\{f = \infty\} = \bigcap_n \{f \geq n\}$. ∎

Proposition 10.7 (10.2.3)

(a) Continuous $f : E \to \mathbb{R}$ are measurable.
(b) If f, g are measurable and real-valued and $a \in \mathbb{R}$, then $f + g$, fg, af are measurable.
(c) If each f_n is measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf f_n$, $\limsup f_n$, and hence $\lim f_n$ where it exists.

Proof. (a) $\{f > a\} = U \cap E$ with U open.

(b) $\{f + g > a\} = \bigcup_{r \in \mathbb{Q}} \{f > r\} \cap \{g > a - r\}$; f^2 is measurable since $\{f^2 > a\} = \{f > \sqrt{a}\} \cup \{f < -\sqrt{a}\}$ for $a \geq 0$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

(c) $\{\sup f_n > a\} = \bigcup \{f_n > a\}$, $\inf f_n = -\sup(-f_n)$, $\limsup f_n = \inf_n \sup_{k \geq n} f_k$. ∎

A **simple function** is $s = \sum_{i=1}^k a_i \chi_{A_i}$ with $a_i \geq 0$ and $A_i \in \mathfrak{M}$; its integral is $\int_E s = \sum a_i m(A_i \cap E)$ (independent of the representation, by passing to a common disjoint refinement and using additivity). For measurable $f : E \to [0, \infty]$,

$$\int_E f = \sup\Big\{ \int_E s : 0 \leq s \leq f,\ s \text{ simple} \Big\}.$$

For simple functions the two definitions agree, and $0 \leq f \leq g$ implies $\int f \leq \int g$.

Proposition 10.8 (10.2.4)

Every measurable $f : E \to [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions.

Proof. Let $A_{n,i} = \{\frac{i-1}{2^n} \leq f < \frac{i}{2^n}\}$ $(1 \leq i \leq n2^n)$, $B_n = \{f \geq n\}$, and $s_n = \sum_i \frac{i-1}{2^n} \chi_{A_{n,i}} + n\chi_{B_n}$. Then $s_n \leq s_{n+1} \leq f$, and $f - s_n < 2^{-n}$ where $f < n$; also $s_n = n \to \infty$ where $f = \infty$. ∎

Theorem 10.9 (10.2.5)

If $A = \bigsqcup A_n$ with $A_n \in \mathfrak{M}$ disjoint and $f : A \to [0, \infty]$ is measurable, then $\int_A f = \sum_n \int_{A_n} f$.

Proof. For simple s, $\int_A s = \sum_n \int_{A_n} s$ by Theorem 10.4; so for simple $s \leq f$, $\int_A s \leq \sum \int_{A_n} f$, and taking the supremum, $\int_A f \leq \sum \int_{A_n} f$. Conversely, given N and ϵ, choose simple $s_k \leq f$ on A_k with $\int_{A_k} s_k > \int_{A_k} f - \epsilon/N$ $(k \leq N)$; $s = \sum_{k \leq N} s_k \chi_{A_k} \leq f$, so $\int_A f \geq \sum_{k \leq N} \int_{A_k} f - \epsilon$. ∎

So $A \mapsto \int_A f$ is a measure. For measurable f, write $f^+ = \max(f, 0)$, $f^- = \max(-f, 0)$; $f \in L^1(E)$ if $\int_E |f| < \infty$, and then $\int_E f = \int_E f^+ - \int_E f^-$; for complex f, $\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f$.

Corollary 10.10 (10.2.6)

Theorem 10.9 holds for every $f \in L^1(A)$, the series converging absolutely.

Proof. Apply Theorem 10.9 to $f^{\pm}$ (real and imaginary parts) and subtract; the series converge because $\sum \int_{A_n} |f| = \int_A |f| < \infty$. ∎

Corollary 10.11 (10.2.7)

A measurable f is in $L^1(E)$ iff $|f| \in L^1(E)$, and then $|\int_E f| \leq \int_E |f|$.

Proof. The first statement is the definition. Choose θ with $e^{-i\theta}\int f = |\int f|$; then $|\int f| = \int \mathrm{Re}(e^{-i\theta}f) \le \int|f|$ (using linearity, Theorem 10.14). ■

Proposition 10.12 (10.2.8)

If $f \ge 0$ is measurable and $\int_E f = 0$, then $f = 0$ almost everywhere.

Proof. With $A_n = \{f \ge 1/n\}$, $\frac{1}{n}m(A_n) \le \int f = 0$, so $m(\{f > 0\}) = m(\bigcup A_n) = 0$. ■

10.3 Convergence theorems

Theorem 10.13 (10.3.1, monotone convergence)

If $0 \le f_1 \le f_2 \le \cdots$ are measurable on E and $f = \lim f_n$, then $\int_E f = \lim \int_E f_n$.

Proof. f is measurable and $\int f_n \le \int f_{n+1} \le \int f$, so $L = \lim \int f_n \le \int f$. Let $0 \le s \le f$ be simple and $0 < c < 1$; $E_n = \{f_n \ge cs\}$ increase to E (where $f(x) = 0$, $x \in E_1$; otherwise $cs(x) < f(x)$). Then $\int f_n \ge \int_{E_n} f_n \ge c\int_{E_n} s \to c\int_E s$ (Theorem 10.9, continuity from below). So $L \ge c\int s$; let $c \to 1$ and take the supremum over s. ■

Theorem 10.14 (10.3.2, linearity)

(a) If $f, g \in L^1(E)$, then $f+g \in L^1(E)$; if $f, g \ge 0$ or $f, g \in L^1(E)$, then $\int(f+g) = \int f + \int g$.
(b) If $f \in L^1(E)$ and $a \in \mathbb{C}$, then $af \in L^1(E)$ and $\int af = a\int f$.

Proof. For simple functions this is immediate. For $f, g \ge 0$ take increasing simple $s_n \to f$, $t_n \to g$ (Proposition 10.8); $s_n + t_n \uparrow f + g$, and Theorem 10.13 gives the formula. Then $|f+g| \le |f| + |g|$ gives integrability. For real $f, g \in L^1$ and $h = f + g$, $h^+ + f^- + g^- = h^- + f^+ + g^+$; integrate and rearrange. Complex functions: real and imaginary parts.

(b) is similar (for $a = i$ the parts are exchanged). ■

Corollary 10.15 (10.3.3)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \sum_n f_n = \sum_n \int_E f_n$.

Proof. Apply Theorem 10.13 to the partial sums, using Theorem 10.14. ■

Theorem 10.16 (10.3.4, Riemann and Lebesgue integrals)

Let $f : [a, b] \to \mathbb{R}$ be bounded.
(a) If f is Riemann integrable, then f is measurable and its Lebesgue integral equals its Riemann integral.
(b) f is Riemann integrable iff the set of its discontinuities has measure 0.

Proof. Choose partitions $P_1 \subset P_2 \subset \cdots$ with $\|P_k\| \to 0$, and let ℓ_k, u_k be the step functions equal to m_i, M_i on the open subintervals of P_k (and to f at partition points). Then $\ell_k \uparrow \ell$, $u_k \downarrow u$, $\ell \le f \le u$, and by monotone convergence (applied to $u_1 - u_k$ and $\ell_k - \ell_1$), $\int \ell = \lim L(f, P_k)$ and $\int u = \lim U(f, P_k)$. Let $P = \bigcup P_k$, a countable set. For $x \notin P$, $u(x)$ and $\ell(x)$ are the limits of sup and inf of f over shrinking neighbourhoods of x, so f is continuous at x iff $u(x) = \ell(x)$.

If f is Riemann integrable, then $U(f, P_k) - L(f, P_k) \to 0$ (proof of Theorem 5.8 of Volume I), so $\int(u - \ell) = 0$ and $u = \ell$ a.e. (Proposition 10.12). Then $f = u$ a.e.; f is measurable (Theorem 10.3(e)) and $\int f = \int u = \lim U(f, P_k)$ is the Riemann integral. This proves (a) and, with the characterization

of continuity, the necessity in (b). Conversely, if the discontinuities have measure 0, then $u = \ell$ a.e., so $\lim(U(f, P_k) - L(f, P_k)) = \int(u - \ell) = 0$, and f is Riemann integrable by Riemann's criterion. ■

Theorem 10.17 (10.3.5, Fatou's lemma)

If $f_n : E \to [0, \infty]$ are measurable, then $\int_E \liminf f_n \le \liminf \int_E f_n$.

Proof. $g_n = \inf_{k \ge n} f_k$ increases to $\liminf f_n$ and $\int g_n \le \int f_k$ for $k \ge n$; apply Theorem 10.13. ■

Theorem 10.18 (10.3.6, Lebesgue's dominated convergence theorem)

Let $f_n : E \to \mathbb{C}$ be measurable with (a) $|f_n| \le g$ for some $g \in L^1(E)$ and (b) $f(x) = \lim f_n(x)$ for each x. Then $f \in L^1(E)$ and $\lim \int_E |f_n - f| = 0$; in particular $\int f_n \to \int f$.

Proof. $|f| \le g$, so $f \in L^1$. Apply Fatou to $2g - |f_n - f| \ge 0$: $\int 2g \le \liminf \int(2g - |f_n - f|) = \int 2g - \limsup \int |f_n - f|$. Hence $\limsup \int |f_n - f| \le 0$. ■

The theorems remain valid when the hypotheses hold only almost everywhere, since sets of measure 0 do not affect integrals.

Theorem 10.19 (10.3.7, Beppo Levi)

If $f_n \in L^1(E)$ and $\sum \int_E |f_n| < \infty$, then $\sum f_n(x)$ converges absolutely for almost every x, its sum f is in $L^1(E)$, and $\int_E f = \sum \int_E f_n$.

Proof. $g = \sum |f_n|$ has $\int g = \sum \int |f_n| < \infty$ (Corollary 10.15), so $g < \infty$ a.e. There the series converges absolutely, with partial sums dominated by g; apply Theorem 10.18. ■

Theorem 10.20 (10.3.8, absolute continuity)

If $f \in L^1(E)$, then for every $\epsilon > 0$ there is $\delta > 0$ with $\int_A |f| < \epsilon$ for all measurable $A \subset E$ with $m(A) < \delta$.

Proof. $\min(|f|, n) \uparrow |f|$, so by monotone convergence choose N with $\int(|f| - \min(|f|, N)) < \epsilon/2$, and $\delta = \epsilon/(2N)$. Then $\int_A |f| \le \epsilon/2 + Nm(A) < \epsilon$. ■

10.4 The spaces L^1 and L^2

Identifying functions that agree a.e., $L^1(E)$ with $\|f\|_1 = \int |f|$ and $L^2(E)$ (measurable f with $\int |f|^2 < \infty$) with $\|f\|_2 = (\int |f|^2)^{1/2}$ are normed spaces; L^2 has the inner product $\langle f, g \rangle = \int f\bar{g}$ ($|f\bar{g}| \le \frac{1}{2}(|f|^2 + |g|^2)$).

Theorem 10.21 (10.4.1)

Every Cauchy sequence in $L^1(E)$ converges.

Proof. By Proposition 7.11 it suffices that $\sum \|f_n\|_1 < \infty$ implies convergence of $\sum f_n$ in L^1. By Theorem 10.19, $f = \sum f_n$ exists a.e. and is in L^1, and $\|f - \sum_{n \le N} f_n\|_1 \le \sum_{n > N} \|f_n\|_1 \to 0$. ■

Theorem 10.22 (10.4.2)

$L^2(E)$ is a Banach space (a Hilbert space).

Proof. Let $\sum\|f_n\|_2 = M < \infty$ and $g = \sum|f_n|$. By the triangle inequality $\|\sum_{n\le N}|f_n|\|_2 \le M$, and monotone convergence gives $\int g^2 \le M^2$, so $g < \infty$ a.e. and $f = \sum f_n$ exists a.e. with $|f| \le g$. Since $|f - \sum_{n\le N} f_n|^2 \le g^2$ and tends to 0 a.e., dominated convergence gives $\|f - \sum_{n\le N} f_n\|_2 \to 0$. Apply Proposition 7.11. ∎

Theorem 10.23 (10.4.3, Riesz–Fischer)

For every $c \in \ell^2(\mathbb{Z})$ there is a periodic $f \in L^2[-\pi,\pi]$ with $\hat{f}(n) = c(n)$ for all n.

Proof. By orthonormality, $s_N = \sum_{|n|\le N} c(n)e^{inx}$ satisfies $\frac{1}{2\pi}\int|s_M - s_N|^2 = \sum_{N<|n|\le M}|c(n)|^2$, so (s_N) is Cauchy in L^2 and converges to some f (Theorem 10.22). By Cauchy–Schwarz, $\hat{f}(n) = \lim_N \widehat{s_N}(n) = c(n)$. ∎

Thus $f \mapsto \hat{f}$ is an isometric isomorphism of $L^2[-\pi,\pi]$ (with $\frac{1}{2\pi}\int$) onto $\ell^2(\mathbb{Z})$, once injectivity is known (Theorem 10.26).

Proposition 10.24 (10.4.4)

If $f \in L^1(E)$ and $\int_A f = 0$ for every measurable $A \subset E$, then $f = 0$ a.e.

Proof. With $A = \{\operatorname{Re} f > 0\}$, $\int_A \operatorname{Re} f = \operatorname{Re}\int_A f = 0$, so $\operatorname{Re} f\chi_A = 0$ a.e. (Proposition 10.12) and $m(A) = 0$. Similarly for $\{\operatorname{Re} f < 0\}$, $\{\operatorname{Im} f > 0\}$, $\{\operatorname{Im} f < 0\}$. ∎

Lemma 10.25 (10.4.5)

For every measurable $E \subset [a,b]$ there are continuous g_n with $|g_n| \le 1$ and $g_n \to \chi_E$ a.e.

Proof. By regularity choose compact K_n and open U_n with $K_n \subset E \subset U_n$ and $m(U_n - K_n) < 2^{-n}$, and let $g_n(x) = \frac{d(x,U_n^c)}{d(x,U_n^c)+d(x,K_n)}$, continuous, equal to 1 on K_n and 0 off U_n. Then $g_n(x) \ne \chi_E(x)$ only for $x \in U_n - K_n$. The set of x lying in infinitely many $U_n - K_n$ is contained in $\bigcup_{n\ge N}(U_n - K_n)$ for every N, so has measure at most 2^{1-N}, i.e. 0. Elsewhere $g_n(x) \to \chi_E(x)$. ∎

Theorem 10.26 (10.4.6)

If $f \in L^1[-\pi,\pi]$ and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z}$, then $f = 0$ a.e.

Proof. By hypothesis $\int fg = 0$ for every trigonometric polynomial g, hence for every continuous periodic g (Fejér, Theorem 9.11, gives uniformly convergent trigonometric polynomials; pass to the limit by dominated convergence). Let $A \subset [-\pi+\eta, \pi-\eta]$ be measurable. Lemma 10.25, with the U_n inside $(-\pi,\pi)$, gives continuous periodic g_n, $|g_n| \le 1$, $g_n \to \chi_A$ a.e.; by dominated convergence $\int_A f = \lim \int f g_n = 0$. Letting $\eta \to 0$, $\int_A f = 0$ for all measurable A, and Proposition 10.24 applies. ∎

Hence two L^1 functions with the same Fourier coefficients agree a.e., and Parseval's identity holds for every $f \in L^2[-\pi,\pi]$.

Lemma 10.27 (10.4.7)

For $f \in L^2(E)$ and $\epsilon > 0$ there is a simple function s (a finite complex combination of characteristic functions of sets of finite measure) with $\|f - s\|_2 < \epsilon$.

Proof. For $f \ge 0$ take increasing simple $s_n \to f$ (Proposition 10.8); $|f - s_n|^2 \le f^2$, so $\|f - s_n\|_2 \to 0$ by dominated convergence, and $\int s_n^2 < \infty$ forces the sets in s_n to have finite measure. In general use $f = (\operatorname{Re} f)^+ - (\operatorname{Re} f)^- + i(\operatorname{Im} f)^+ - i(\operatorname{Im} f)^-$. ∎

Theorem 10.28 (10.4.8)

For $f \in L^2[a,b]$ and $\epsilon > 0$ there is $g \in C[a,b]$ with $\|f - g\|_2 < \epsilon$.

Proof. By Lemma 10.27 it suffices to approximate χ_A, $A \subset [a,b]$ measurable. With g_n from Lemma 10.25, $|g_n - \chi_A|^2 \le 4$ and tends to 0 a.e., so $\|g_n - \chi_A\|_2 \to 0$ by dominated convergence on the finite interval. ■

Theorem 10.29 (10.4.9)

For $f \in L^1(E)$ and $\epsilon > 0$ there is a continuous g on $\mathbb{R}$ vanishing outside a bounded interval with $\|f - g\|_1 < \epsilon$ (norm taken over E).

Proof. By dominated convergence $\|f - f\chi_{[-N,N]}\|_1 \to 0$; approximate $f\chi_{[-N,N]}$ in L^1 by simple functions as in Lemma 10.27; approximate each $\chi_{A\cap E}$, $A \subset [-N,N]$, by g_n from Lemma 10.25 (with $U_n \subset (-N-1, N+1)$), using dominated convergence. ■

In particular L^1 and L^2 are the completions of the spaces $\mathcal{R}^1$, $\mathcal{R}^2$ of Chapter 7.

10.5 The Poisson summation formula

For $g \in L^1(\mathbb{R})$ let $G(x) = \sum_{n\in\mathbb{Z}} g(x + 2\pi n)$.

Theorem 10.30 (10.5.1)

The series defining G converges absolutely for almost every x, $G \in L^1[-\pi,\pi]$, and

$$\hat{G}(n) = \frac{1}{2\pi}\int_{-\infty}^{\infty} g(x)e^{-inx}\,dx, \qquad n \in \mathbb{Z}.$$

Proof. Integrals over $\mathbb{R}$ are translation invariant (Proposition 10.2(c) for sets, hence for simple functions and by monotone convergence for all $f \ge 0$). By Corollary 10.15, $\int_{-\pi}^{\pi}\sum_n |g(x+2\pi n)|\,dx = \sum_n \int_{-\pi+2\pi n}^{\pi+2\pi n}|g| = \int_{\mathbb{R}}|g| < \infty$. So the series converges absolutely a.e. and $G \in L^1$. By Theorem 10.19,

$$\hat{G}(n) = \frac{1}{2\pi}\sum_k \int_{-\pi}^{\pi} g(x+2\pi k)e^{-inx}dx = \frac{1}{2\pi}\sum_k \int_{-\pi+2\pi k}^{\pi+2\pi k} g(y)e^{-iny}dy = \frac{1}{2\pi}\int_{\mathbb{R}} ge^{-iny}dy.$$

■

If moreover the Fourier series of G converges to $G(0)$ at 0 (for example if g is smooth and rapidly decreasing), this gives the **Poisson summation formula** $\sum_n g(2\pi n) = \frac{1}{2\pi}\sum_n \int_{\mathbb{R}} g(x)e^{-inx}dx$. Applied to $g(x) = e^{-ax^2}$ it yields the transformation formula for the theta function $\theta(x) = \sum_n e^{-\pi n^2 x}$: $\theta(1/x) = \sqrt{x}\,\theta(x)$.